



Murriyang Data Access: Pulsar Data

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This document describes the ways in which pulsar data taken by the Parkes radio telescope can be accessed.

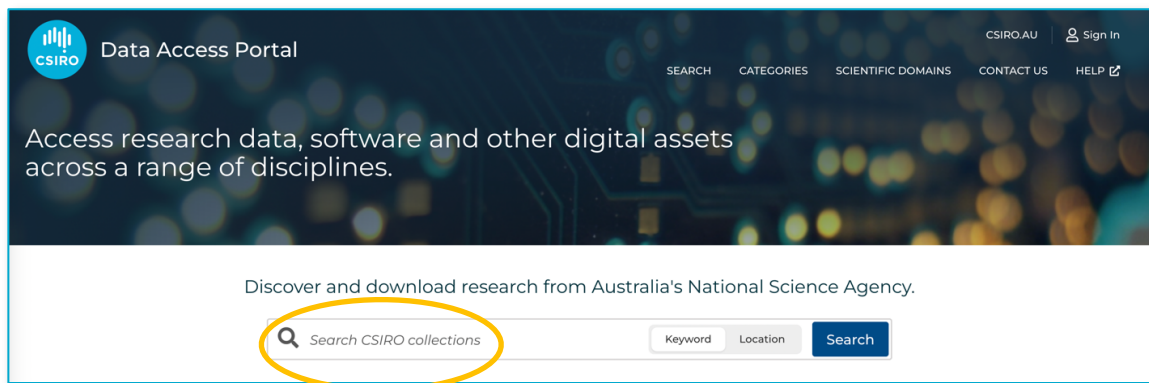
Pulsar data: Accessing data from CSIRO's Data Access Portal (DAP)

CSIRO's DAP is the most comprehensive archive for Parkes pulsar data available globally, with a volume of ~6 Petabytes and containing observations dating from the early 1990's to today.

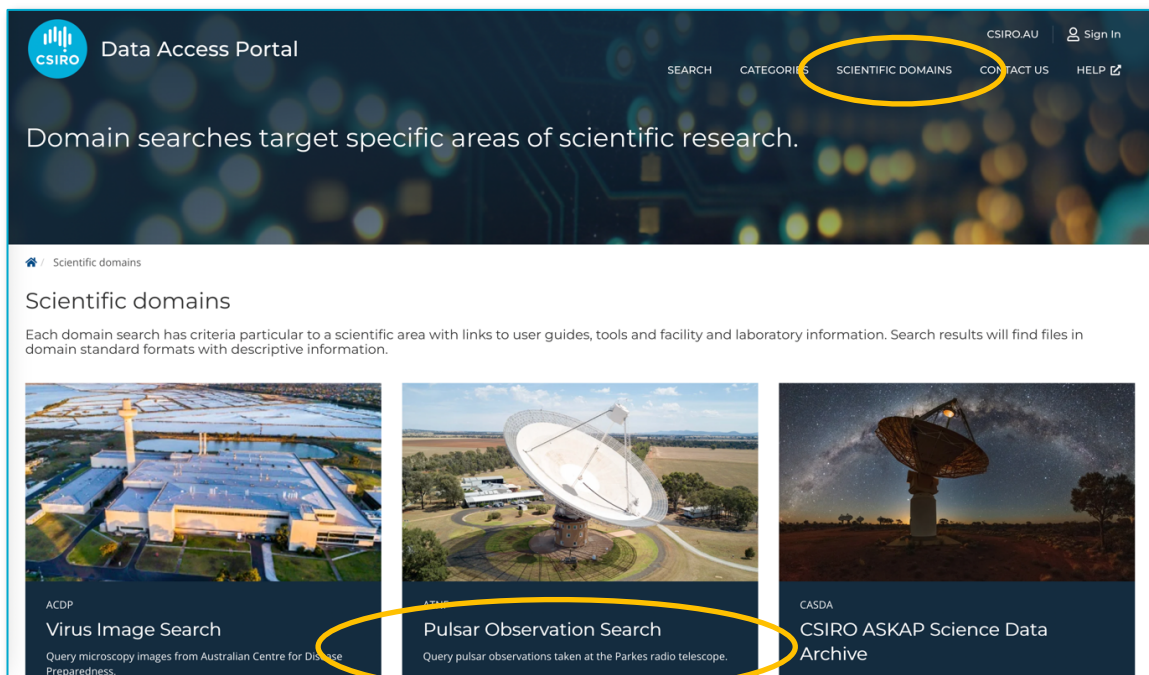
The preferred method for pulsar data access given the large volumes is via an S3 endpoint. A step-by-step guide on how to access data from an S3 endpoint is presented here.

STEP 1: Conduct a search query

- Navigate to <https://data.csiro.au>
- Search for your data either:
 - a) by conducting a keyword search, for example by Project ID:



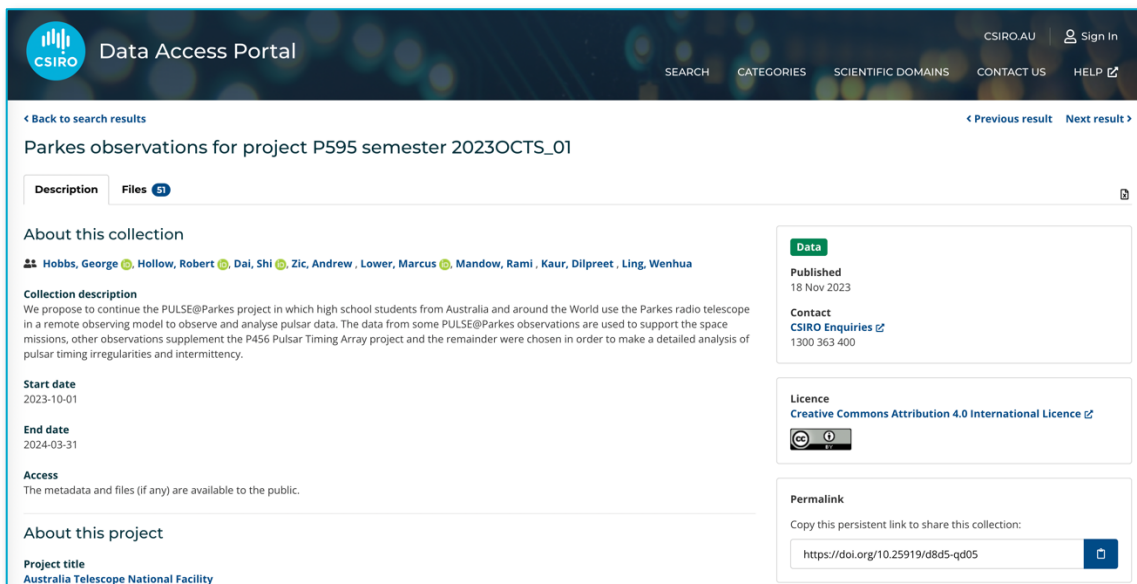
Or b), by conducting a specific 'ATNF Pulsar Observation Search', for example by source name, position, MJD, backend or filename:



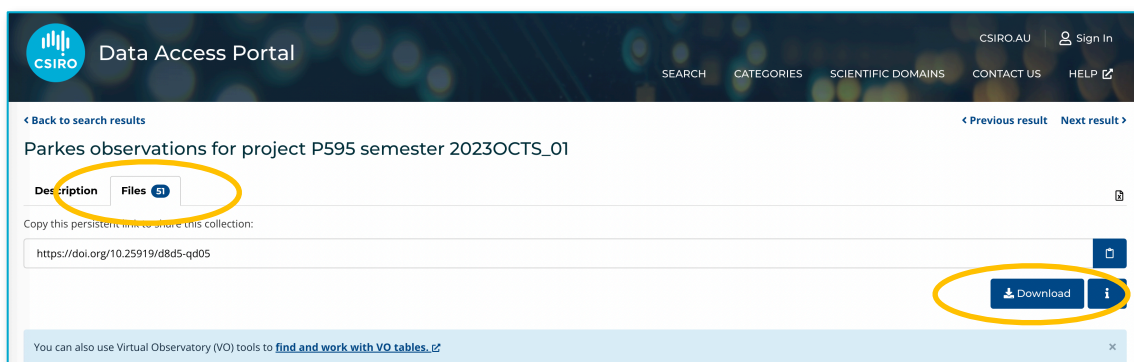
STEP 2: Request access to a data collection

Please note: if your data are spread across multiple collections, you currently need to request each collection separately.

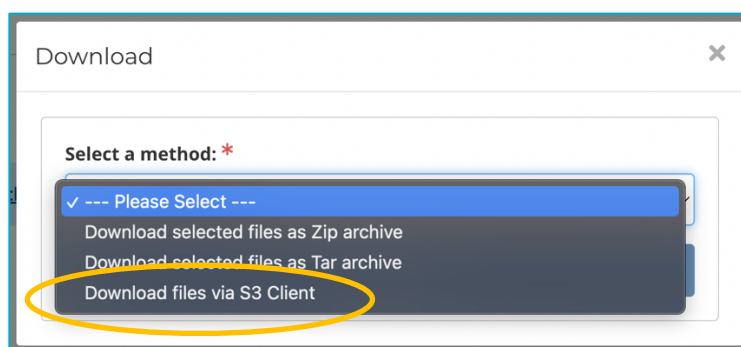
In the following example, the keyword search query for Project ID 'P595' returned a list of 187 collections, and the '2023APRS_01' collection was selected:



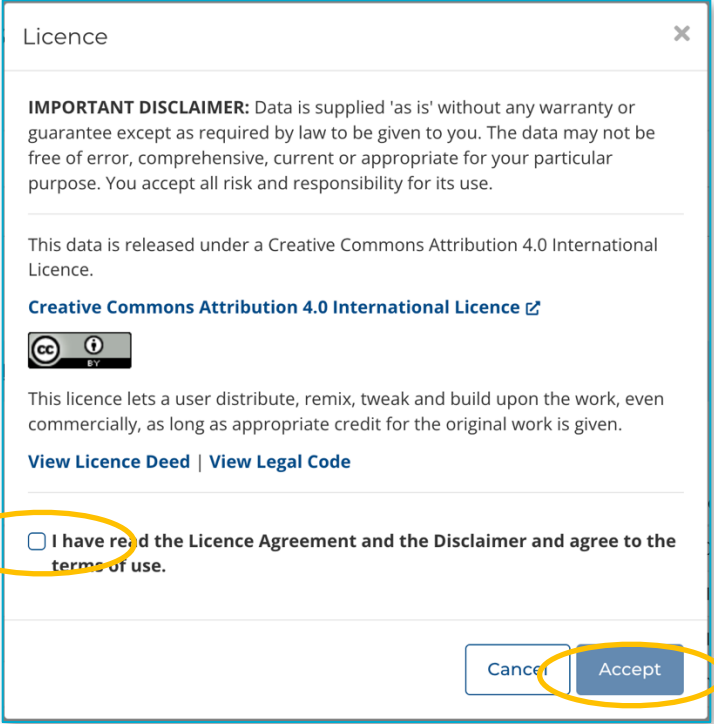
- In the example above, P595 collections are not embargoed, so anyone has access. If you are on the OPAL proposal for an embargoed collection, log in with your OPAL credentials to request the collection.
- Click the 'Files' tab and then 'Download':



- Select 'Download files via S3 Client', enter your email address and click 'Request files':



- Agree to the licence conditions:



The 'Licence' dialog box contains the following text:

IMPORTANT DISCLAIMER: Data is supplied 'as is' without any warranty or guarantee except as required by law to be given to you. The data may not be free of error, comprehensive, current or appropriate for your particular purpose. You accept all risk and responsibility for its use.

This data is released under a Creative Commons Attribution 4.0 International Licence.

[Creative Commons Attribution 4.0 International Licence](#)



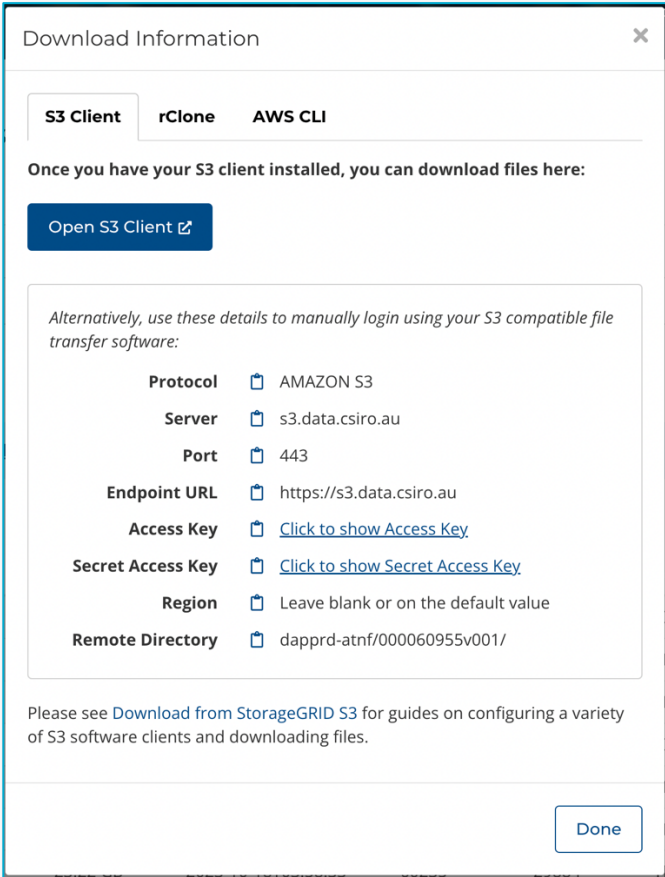
This licence lets a user distribute, remix, tweak and build upon the work, even commercially, as long as appropriate credit for the original work is given.

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☐ I have read the Licence Agreement and the Disclaimer and agree to the terms of use.

Buttons: Cancel, Accept

On receipt of the request, you will see this pop-up dialogue where you can choose your S3 client and receive relevant help information:



The 'Download Information' dialog box contains the following content:

S3 Client | rClone | AWS CLI

Once you have your S3 client installed, you can download files here:

[Open S3 Client](#)

Alternatively, use these details to manually login using your S3 compatible file transfer software:

Protocol	AMAZON S3
Server	s3.data.csiro.au
Port	443
Endpoint URL	https://s3.data.csiro.au
Access Key	Click to show Access Key
Secret Access Key	Click to show Secret Access Key
Region	Leave blank or on the default value
Remote Directory	dapprd-atnf/000060955v001/

Please see [Download from StorageGRID S3](#) for guides on configuring a variety of S3 software clients and downloading files.

Done

STEP 3: Download your data

- Choose from the selection of S3 clients as shown above, and follow the instructions supplied, and/or refer to the guide for further information.

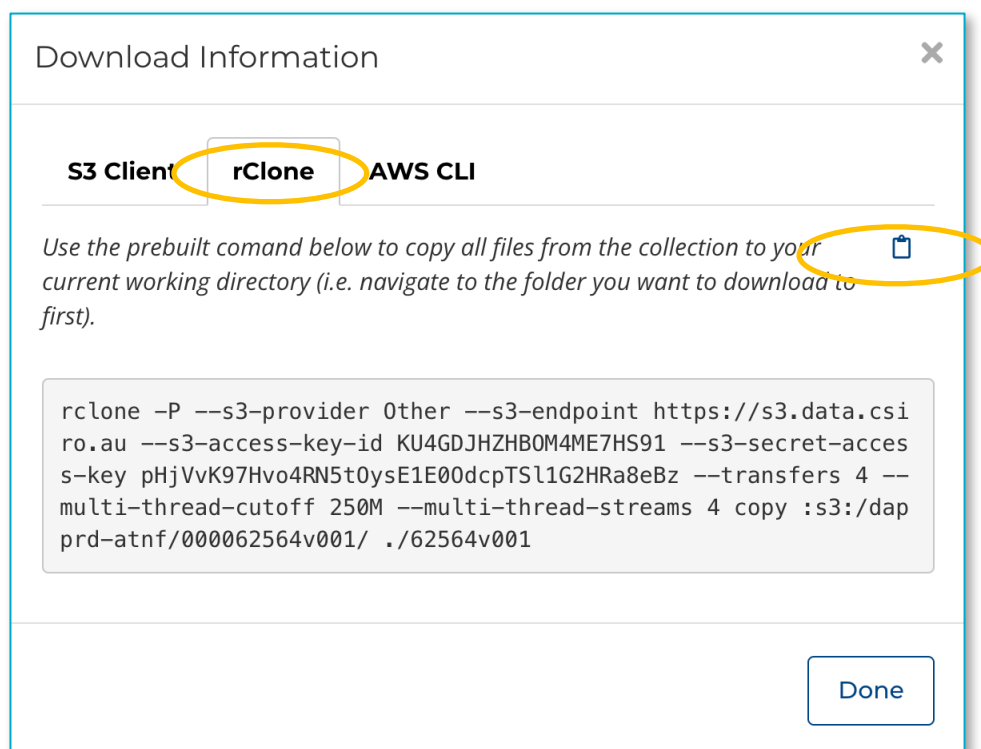
ATNF Computer Account Holders

An ATNF computer account holder can download data using *rclone* to pre-booked disk space on an ATNF machine.

- Log on to an ATNF machine
- Check for booked disk space with:

```
bookings -u your_ident
```

- To request disk space, please email bookings@atnf.csiro.au with your requirements
- Once you have been allocated disk space, data can be downloaded by copying the *rclone* command specified in the pop-up dialogue 'rClone' tab, and then pasting it into your terminal:



Command-line data access tools

The ATNF provides data query and access tools – use of these is the preferred method for accessing large data volumes and/or retrieving a sub-set of observations.

On ATNF servers, the following tools are available:

- `dap-query.sh` - query the DAP database
- `get-dap-psr-data.sh` - download unembargoed DAP data specified by a `dap-query.sh` result set
- `get-dap-psr-collection.sh` - download embargoed/unembargoed DAP collections or specify files to download

Need help with data access?

Information about the ATNF Data Archives is available at:

<https://www.atnf.csiro.au/observers/data/index.html>

Contact ATNF data support at:

atnf-datasup@csiro.au

Please contact Lawrence.Toomey@csiro.au for further information.